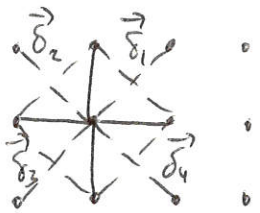
P2a) Spinwave analysis of $J_1 - J_2$ model

$d=2$

Suppose we add $J_2 > 0$: it connects spins on the same sublattice A with A, B with B



$\Rightarrow$ among A spins m leads to:

$$\delta H_A = \frac{1}{2} J_2 \sum_{\substack{nn' \\ n'=n+\delta}} S_n^z S_{n'}^z + \frac{1}{2}\left(S_n^+ S_{n'}^- + S_n^- S_{n'}^+\right)$$

$$= \frac{1}{2} J_2 \sum_{nn'} (S - a_n^+ a_n)(S - a_{n'}^+ a_{n'}) + \frac{1}{2}(2S)\left(a_n a_{n'}^+ + a_n^+ a_{n'}\right)$$

$$= \frac{1}{2} J_2 \sum_{n\delta} S^2 - S\left(a_n^+ a_n + a_{n+\delta}^+ a_{n+\delta} - a_{n+\delta}^+ a_n - a_n^+ a_{n+\delta}\right) + O(1)$$

$$= \frac{1}{2} J_2 S^2 \sum_{n\delta}(1) - \frac{1}{2} J_2 S \sum_{\delta} \sum_{k} \left(a_k^+ a_k + a_k^+ a_k - a_k^+ a_k e^{i\bar{k}\bar{\delta}}\right.$$

$$\left. - a_k^+ a_k e^{-ik\delta}\right) = \frac{1}{2} J_2 S^2 \frac{N}{2} \cdot 4 \quad +$$

$$- \frac{1}{2} J_2 S \sum_k \sum_\delta 2\left(1 - \cos \bar{k}\bar{\delta}\right) a_k^+ a_k$$

$$\sum_{\bar{\delta}} \left(1 - \cos \bar{k}\bar{\delta}\right) = \underset{\delta_1, \delta_3}{2\left(1 - \cos(k_x + k_y)a\right)} + \underset{\delta_2, \delta_4}{2\left(1 - \cos(k_x - k_y)a\right)}$$

$$= 4 - 2 \cdot 2 \cos k_x a \cdot \cos k_y a = 4\left(1 - \cos k_x a \cos k_y a\right)$$

$$\delta H_A = J_2 S^2 N - \sum_k J_2 S \cdot 4\left(1 - \cos k_x a \cos k_y a\right) a_k^+ a_k$$

Clearly δH_B is of the same form:

$$\delta H_B = J_2 S^2 N - \sum_k J_2 S \cdot 4\left(1 - \cos k_x a \cos k_y a\right) b_k^+ b_k$$

Then full harmonic hamiltonian of the 2d problem is

$$H^{(2)} = \sum_k \left[4J_1 S - 4J_2 S\left(1-\cos k_x \cos k_y\right)\right]\left(a_k^+ a_k + b_k^+ b_k\right)$$

$$+ 4J_1 S \cdot \frac{1}{2}\left(\cos k_x + \cos k_y\right)\left(a_k^+ b_{-k}^+ + b_k a_{-k}\right)$$

$$\omega_k = \sqrt{A_k^2 - B_k^2} = \left(4J_1 S\right)\sqrt{\left(1-\alpha\left(1-\cos k_x \cos k_y\right)\right)^2 - \left(\frac{1}{2}\left(\cos k_x + \cos k_y\right)\right)^2}$$

$$\alpha \equiv \frac{J_2}{J_1}$$

Then $$\langle S_n \rangle = S\left\{1 - \frac{1}{2S}\frac{2}{N}\sum_k\left(\frac{A_k}{\omega_k} - 1\right)\right\} =$$

$$= S\left\{1 - \frac{1}{2S}\frac{2}{N}\sum_k\left(\frac{4J_1 S\left[1-\alpha\left(1-\cos k_x \cos k_y\right)\right]}{4J_1 S\sqrt{}} - 1\right)\right\}$$

As before $\frac{2}{N}\sum(\quad) = \frac{2}{(2\pi)^2}\int_{MBZ} dx\,dy\,(\quad)$

$$\frac{\langle S_n \rangle}{S} = 1 - \frac{1}{2S}\left[\frac{2\cdot 4}{(2\pi)^2}\int_0^{\pi} dx \int_0^{\pi - x} dy\, \frac{1-\alpha\left(1-\cos x \cos y\right)}{\sqrt{\left(1-\alpha\left(1-\cos x \cos y\right)\right)^2 - \frac{1}{4}\left(\cos x + \cos y\right)^2}} - 1\right]$$

For $S = \frac{1}{2}$ magnetic long range order (LRO) disappears when $\frac{\langle S_n \rangle}{S} = 0$

which happens when $[\ ] = 1 \Rightarrow$ critical α_c when

$$J_2(\alpha_c) = 2$$

here $$J_2(\alpha) = \frac{2}{\pi^2}\int_0^{\pi} dx \int_0^{\pi - x} dy \; \frac{1-\alpha(1-\cos x\cos y)}{\sqrt{\left[1-\alpha(1-\cos x\cos y)\right]^2 - \frac{1}{4}(\cos x+\cos y)^2}}$$

Numerically we find $\alpha_c \simeq 0.38$.

At the same time $\frac{1}{S}$ correction to the ground state energy remains smooth — it does not reflect the loss of magnetic order because it is determined by short-range correlations between spins.

$$E = -J_1 S^2 \frac{N}{2}\cdot 4 + 2\cdot\frac{1}{2}J_2 S^2 \frac{N}{2}\cdot 4$$

$$+ \sum_k (\omega_k - A_k)$$

$$A_k = 4J_1 S - 4J_2 S\,(1-\cos k_x \cos k_y)$$

$$\Rightarrow \sum_k A_k = \frac{N}{2}\left(4J_1 S - 4J_2 S\right)$$

$$E = +\frac{N}{2}\cdot 4\left(-J_1 S^2 + J_2 S^2\right) - \frac{N}{2}\cdot 4 S\left(J_1 - J_2\right)$$

$$+\frac{N}{2}\cdot 4\left(\frac{2}{N}\,\frac{1}{4}\sum_k \omega_k\right) =$$

$$= 2N\,S(S+1)\left(J_2 - J_1\right) + 2N\cdot\frac{2}{N}\sum_k \frac{4J_1 S}{4}\sqrt{}$$

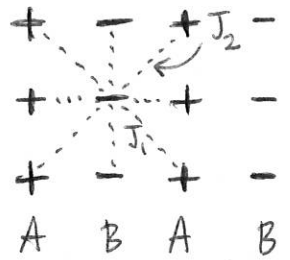
<u>Energy per spin</u>

$$\frac{E}{N} = 2S(S+1)(J_2 - J_1) + 2S\,\frac{2}{(2\pi)^2}\int_{MBZ}\sqrt{\left[1-\alpha(1-\cos x\cos y)\right]^2 - \frac{1}{4}\gamma_k^2}$$

P2b

$$\mu = \frac{J_2}{J_1} > \frac{1}{2}$$

Classical structure is minimized by $\vec{Q} = \left(\frac{\pi}{a}, 0\right)$



$\underline{\underline{A}}$ ⊕ sites: $S^z = S - a^+ a$

$S^+ = \sqrt{2S}\, a \;,\; S^- = \sqrt{2S}\, a^+$

$\underline{\underline{B}}$ ⊖ sites: $S^z = -S + b^+ b$

$S^+ = \sqrt{2S}\, b^+ \;,\; S^- = \sqrt{2S}\, b$

$$H_1 = \frac{1}{2} J_1 \sum_{n g} \vec{S}_n \vec{S}_{n+g} =$$

$$= \frac{1}{2} J_1 \Big(\sum_{\substack{n \in A \\ g_x = \pm \hat{x}}} + \sum_{\substack{n \in B \\ g_x = \pm \hat{x}}} + \sum_{\substack{n \in A \\ g_y = \pm \hat{y}}} + \sum_{\substack{n \in B \\ g_y = \pm \hat{y}}} \Big) \vec{S}_n \vec{S}_{n+g}$$

$$= J_1 \sum_{\substack{n=A \\ g_x}} \Big\{ (S - a_n^+ a_n)(-S + b_{n+g_x}^+ b_{n+g_x}) + \frac{1}{2}(2S)(a_n b_{n+g_x} + a_n^+ b_{n+g_x}^+) \Big\}$$

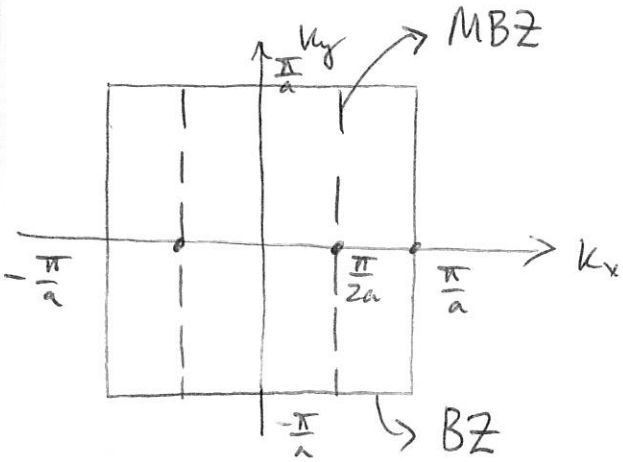
$$+ \frac{1}{2} J_1 \sum_{\substack{n=A \\ g_y}} \Big\{ (S - a_n^+ a_n)(S - a_{n+g_y}^+ a_{n+g_y}) + S(a_n a_{n+g_y}^+ + a_n^+ a_{n+g_y} \Big\}$$

$$+ \frac{1}{2} J_1 \sum_{\substack{n=B \\ g_y}} \Big\{ (-S + b_n^+ b_n)(-S + b_{n+g_y}^+ b_{n+g_y}) + S(b_n^+ b_{n+g_y} + b_n b_{n+g_y}^+ \Big\}$$

$$= -J_1 S^2 \sum_{\substack{n=A \\ g_x}} 1 + 2 \cdot \frac{1}{2} J_1 S^2 \sum_{\substack{n=A \\ g_y}} 1 +$$

$$+ J_1 S \sum_{\substack{n=A \\ g_x}} \left(b_{n+g_x}^+ b_{n+g_x} + a_n^+ a_n + a_n^+ b_{n+g_x}^+ + a_n b_{n+g_x} \right)$$

$$+ \frac{1}{2} J_1 S \sum_{\substack{n=A \\ g_y}} \Big\{ a_n^+ a_{n+g_y} + a_{n+g_y}^+ a_n - a_n^+ a_n - a_{n+g_y}^+ a_{n+g_y} + b_n^+ b_{n+g_y} + b_{n+g_y}^+ b_n - b_n^+ b_n - b_{n+g_y}^+ b_{n+g_y} \Big\} + O\left(\frac{1}{S}\right)$$



Bril. zone for a(b) bosons is $\frac{1}{2}$ of the full BZ:

$$\begin{cases} k_x \in \left[-\frac{\pi}{2a}, \frac{\pi}{2a}\right] \\ k_y \in \left[-\frac{\pi}{a}, \frac{\pi}{a}\right] \end{cases}$$

$$a_n = \sqrt{\frac{2}{N}} \sum_{k=MBZ} e^{-i\vec{k}\vec{R}_n} a_k$$

$$H_1 = J_1 S \sum_{k g_x} \left(b_k^+ b_k + a_k^+ a_k + a_k^+ b_{-k}^+ e^{-i\vec{k}\vec{g}_x} + a_k b_{-k} e^{ikg_x} \right)$$

$$+ \frac{1}{2} J_1 S \sum_{k g_y} \Big\{ a_k^+ a_k \left(e^{i\bar{k}\bar{g}_y} + e^{-i\bar{k}\bar{g}_y} \right) - 2 a_k^+ a_k$$

$$+ b_k^+ b_k \, 2\cos(\bar{k}\cdot\bar{g}_y) - 2 b_k^+ b_k \Big\} =$$

$$= J_1 S \sum_k \Big\{ a_k^+ a_k \left(\sum_{g_y} (\cos k_y a - 1) + \sum_{g_x} 1 \right) +$$

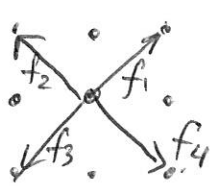
$$+ b_k^+ b_k \left(\sum_{g_y} (\cos k_y a - 1) + \sum_{g_x} 1 \right) + a_k^+ b_{-k}^+ \sum_{g_x} e^{-ik_x a}$$

$$+ a_k b_{-k} \sum_{g_x} e^{ik_x a} \Big\}$$

$$= 2 J_1 S \sum_k \Big\{ \cos k_y a \left(a_k^+ a_k + b_k^+ b_k \right) + \cos k_x a \left(a_k^+ b_{-k}^+ + a_k b_{-k} \right) \Big\}$$

Next $H_2 = \frac{1}{2} J_2 \sum_{n f} \vec{S}_n \vec{S}_{n+f}$

where $\vec{f}$ denotes next-nearest-neighbor bonds



Observe that J_2 connects A and B sites.

$$H_2 = \frac{1}{2} J_2 \sum_{\substack{n=A \\ f}} \Big\{ (S - a_n^+ a_n)(-S + b_{n+f}^+ b_{n+f}) +$$

$$+ \frac{1}{2}(2S)(a_n b_{n+f} + a_n^+ b_{n+f}^+) \Big\} +$$

$$+ \frac{1}{2} J_2 \sum_{\substack{n=B \\ f}} \Big\{ (-S + b_n^+ b_n)(S - a_{n+f}^+ a_{n+f}) + \frac{1}{2}(2S) \cdot$$

$$\cdot (b_n^+ a_{n+f}^+ + b_n a_{n+f}) \Big\} =$$

$$= -\frac{1}{2} J_2 S^2 \Big(\sum_{\substack{n=A \\ f}} 1 + \sum_{\substack{n=B \\ f}} \Big) + J_2 S \sum_{\substack{n=A \\ f}} \Big\{ a_n^+ a_n + b_{n+f}^+ b_{n+f} +$$

$$+ a_n b_{n+f} + a_n^+ b_{n+f}^+ \Big\} = \underbrace{-4 J_2 S^2 \cdot \frac{N}{2}}_{\text{classical energy}}$$

$$+ J_2 S \sum_{kf} \Big\{ a_k^+ a_k + b_k^+ b_k + a_k^+ b_{-k}^+ e^{-i\vec{k}\vec{f}} + a_k b_{-k} e^{i\vec{k}\vec{f}} \Big\}$$

where $t_k \equiv \sum_f e^{i\vec{k}\vec{f}} = \underset{(f_1)}{e^{i(k_x+k_y)a}} + \underset{(f_3)}{e^{-i(k_x+k_y)}} + \underset{(f_2)}{e^{-i(k_x-k_y)a}} + \underset{(f_4)}{e^{i(k_x-k_y)a}}$

$$= 2\cos(k_x+k_y)a + 2\cos(k_x-k_y)a = 4\cos(k_x a)\cos(k_y a)$$

Collect together $H_1^{(2)}$ and $H_2^{(2)}$:

$$H^{(2)} = S \sum_k \Big\{ (4J_2 + 2J_1 \cos k_y a)(a_k^+ a_k + b_k^+ b_k)$$

$$+ \big(4J_2 \cos(k_x a)\cos(k_y a) + 2J_1 \cos(k_x a)\big)(a_k^+ b_{-k}^+ + a_k b_{-k}) \Big\}$$

$\mu = \frac{J_2}{J_1}$

$$H^{(2)} = 2SJ_1 \sum_k \Big\{ \overbrace{\left(2\mu + \cos(k_y a)\right)}^{A_k} \left(a_k^+ a_k + b_k^+ b_{+k}\right)$$

$$+ \underbrace{\cos k_x a \left(2\mu \cos(k_y a) + 1\right)}_{B_k} \left(a_k^+ b_{-k}^+ + a_k b_{-k}\right) \Big\}$$

By diagonalizing we find excitations with spectrum

$$\omega_k = \sqrt{A_k^2 - B_k^2} = 2SJ_1 \left[(2\mu + \cos k_y a)^2 - \cos^2 k_x a \left(2\mu \cos k_y a + 1\right)^2 \right]^{1/2}$$

Then average spin follows from:

$$\langle S_n^z \rangle = S \left\{ 1 - \frac{1}{2S} \frac{2}{N} \sum_{k=MBZ} \left(\frac{A_k}{\omega_k} - 1 \right) \right\}$$

$$K_b(\mu) = \frac{2}{N} \sum_k \frac{A_k}{\omega_k} = \frac{2}{(2\pi)^2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} dx \int_{-\pi}^{\pi} dy \frac{2\mu + \cos y}{\sqrt{(2\mu + \cos y)^2 - \cos^2 x (2\mu \cos y + 1)^2}}$$

$$= \frac{2}{\pi^2} \int_0^{\pi/2} dx \int_0^{\pi} dy \frac{2\mu + \cos y}{\left[(2\mu + \cos y)^2 - \cos^2 x (2\mu \cos y + 1)^2 \right]^{1/2}}$$

Thus $\langle S_n^z \rangle = S \left(1 - \frac{1}{2S} \left[K_b(\mu) - 1 \right] \right)$

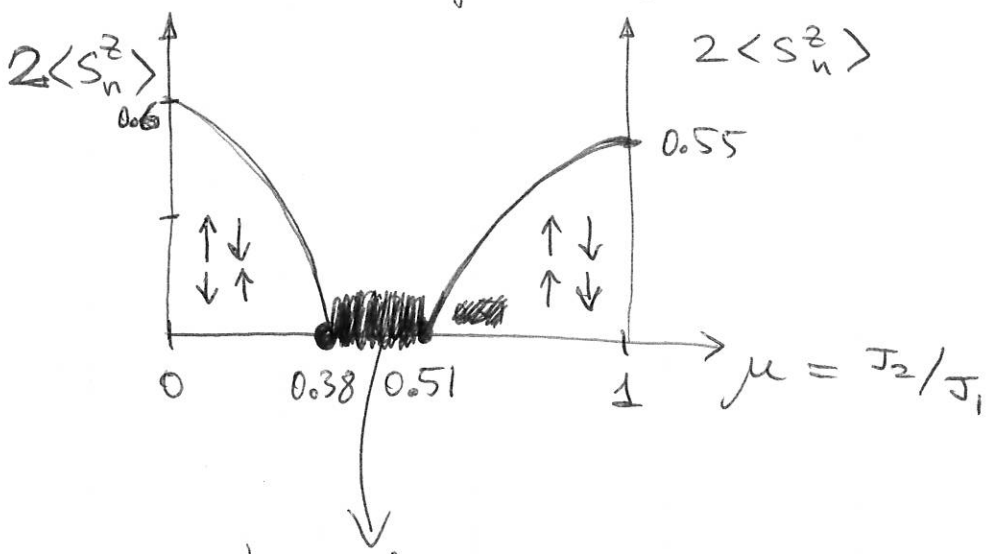
For $S = \frac{1}{2}$ $\langle S_n^z \rangle = 0$ when $K_b(\mu) - 1 = 1$

$\Rightarrow$ look for $K_b(\mu_c) = 2$.

Numerically find that $\mu_c = \left(\frac{J_2}{J_1} \right)_c \approx 0.513$

P2c

Thus there is finite window



here linear spin waves predict disordered ground state: quantum fluctuations destroy classical magnetic order for $0.38 < \mu < 0.51$

(assuming that large-S approximation makes sense even for $S=\frac{1}{2}$. Which is not guaranted but seems likely: extensive numerical studies for $S=\frac{1}{2}$ quantum Heisenberg antiferromagnet ($J_1 - J_2$ model) indicates <u>non-classical</u> ground state in the window $0.38 \leq \mu \leq 0.60$!)

see Sushkov, Oitmaa, Weihong Phys Rev B <u>63</u> 104420 (2001)

and references therein.

Most recent surprise: the ground state is actually <u>topological Z_2 spin liquid</u>,
(Summer 2012!)

See Hong-Chen Jiang, Hong Yao, and Leon Balents, Phys Rev B <u>86</u> 024424 (2012)